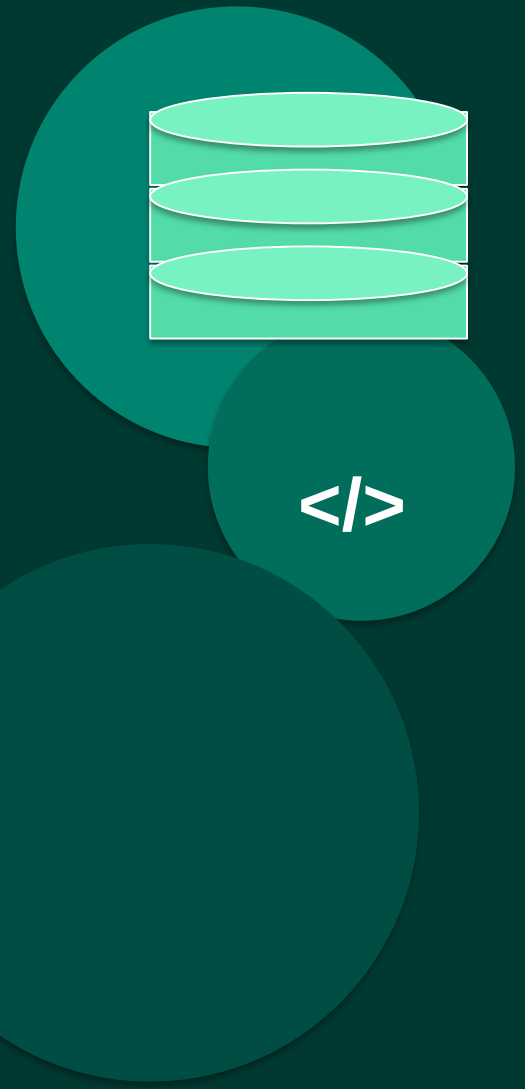
SQL INTRODUCTION

Database Management System (DBMS)

CLASS 01



Learn • Practice • Build Your Skills

Learning Objectives

After completing this class, students will be able to:

- Define data, database, DBMS, and SQL.
- Explain why SQL is used with relational databases.
- Identify the main categories of SQL commands.
- Create a database and a table using SQL.
- Insert, retrieve, update, and delete records.

What Is SQL?

Definition

SQL stands for Structured Query Language. It is a standard language for communicating with relational database management systems (RDBMS).

What can SQL do?

- Create database objects
- Store and retrieve information
- Update existing records
- Delete records when required

Popular SQL database systems

- MySQL — common in web apps
- Microsoft SQL Server — enterprise systems
- PostgreSQL — open-source RDBMS

Basic Database Concepts

Term	Meaning	Example
Data	Raw facts or values	101, Rahim, 85
Database	Organized collection of data	Student database
DBMS	Software for managing databases	MySQL
Table	Data arranged in rows and columns	Students
Row / Record	One complete entry	One student
Column / Field	An attribute of the record	Student name

Example: Students table

Student_ID	Name	Department	Marks
101	Rahim	CSE	85
102	Karim	EEE	78
103	Sumaiya	CSE	92

Rows = records

Columns = attributes

Types of SQL Commands

Category	Full form	Purpose	Examples
DDL	Data Definition Language	Defines structure	CREATE, ALTER, DROP
DML	Data Manipulation Language	Changes data	INSERT, UPDATE, DELETE
DQL	Data Query Language	Retrieves data	SELECT
DCL	Data Control Language	Manages permissions	GRANT, REVOKE
TCL	Transaction Control Language	Manages transactions	COMMIT, ROLLBACK

Note: Some textbooks classify SELECT under DML rather than listing DQL separately.

Why Do We Need SQL? — CRUD

CREATE	READ	INSERT	UPDATE	DELETE
Create database objects	Retrieve information	Add new records	Modify records	Remove records
CREATE TABLE	**SELECT**	**INSERT INTO**	**UPDATE**	**DELETE**

These operations form the foundation of everyday database applications.

Practical 1: Create a Database and Table

Step 1 — Create and select a database

```
CREATE DATABASE UniversityDB;
```

Step 2 — Create the Students table

```
CREATE TABLE Students (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Department VARCHAR(20),
  Marks INT
);
```

Key terms

INT: stores integer values

VARCHAR(n): variable-length text

PRIMARY KEY: uniquely identifies each row

Run the queries

Use MySQL Workbench or phpMyAdmin to execute the examples.

Practical 2: Insert and Retrieve Records

Step 3 — Insert records

```
INSERT INTO Students
  (Student_ID, Name, Department, Marks)
VALUES
  (101, 'Rahim', 'CSE', 85),
  (102, 'Karim', 'EEE', 78),
  (103, 'Sumaiya', 'CSE', 92);
```

Step 4 — Retrieve records

```
SELECT * FROM Students;
```

Select specific columns

```
SELECT Name, Marks
FROM Students;
```

The asterisk (*) means “all columns.”

Practical 3: Filter, Update, and Delete

Filter records with WHERE

```
SELECT * FROM Students
WHERE Department = 'CSE';
```

Update a record

```
UPDATE Students
SET Marks = 88
WHERE Student_ID = 101;
```

Delete a record

```
DELETE FROM Students
WHERE Student_ID = 102;
```

Important safety rule

Always check the WHERE condition before UPDATE or DELETE. Without WHERE, all rows may be affected.

Try SQL Yourself

Practice question

Write a query to display the names and marks of students who scored more than 80.

Table: Students

Columns: Student_ID, Name, Department, Marks

One correct answer

```
SELECT Name, Marks
FROM Students
WHERE Marks > 80;
```

Expected students:

Rahim (85) and Sumaiya (92)

Try changing the condition to Marks < 80 or Department = 'EEE'.

Summary

- SQL is used to work with relational databases.
- A database stores organized data, often in tables.
- CREATE defines database objects; INSERT adds records.
- SELECT retrieves records; WHERE filters results.
- UPDATE modifies records; DELETE removes records.
- Use WHERE carefully when changing or deleting data.

Remember

Practice with small tables first, then build larger database applications.

Homework and Next Class

Homework

- Define SQL and explain its importance.
- Differentiate database and DBMS.
- Write a query to show EEE students.
- Write a query to show marks below 80.
- Create an Employees table with Employee_ID, Name, Designation, and Salary.

Next class

SQL Constraints

PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, and DEFAULT

Learning goal

Design tables that maintain accurate and consistent data.